White Test 1

Complete Solutions Guide

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November 23, 2025

1 Node1 Configuration Solutions

1. **Set multi-user target as default and reboot**

```
systemctl set-default multi-user.target
reboot
systemctl isolate graphical.target
```

2. **Configure Host Name, IP Address, Gateway and DNS**

```
# Set hostname node1.tekup.com
hostnamectl set-hostname node1.tekup.com

# Configure network (using nmcli) , this is the hard and not recommended way of
configuring the network
nmcli con mod eth0 ipv4.addresses 172.24.40.40/24
nmcli con mod eth0 ipv4.gateway 172.24.40.1
nmcli con mod eth0 ipv4.dns 172.24.40.1
nmcli con mod eth0 ipv4.method manual
nmcli con up eth0

# Configure network and hostname using ***** nmtui ******
```

3. **Configure repos AppStream and BaseOS**

```
# Create AppStream repo
cat > /etc/yum.repos.d/appstream.repo << EOF
[appstream]
name=AppStream
baseurl=http://centos.anexia.at/centos-stream/9-stream/AppStream/x86_64/os/
enabled=1
gpgcheck=0
EOF

# Create BaseOS repo
cat > /etc/yum.repos.d/baseos.repo << EOF
[baseos]
name=BaseOS
baseurl=https://centos.anexia.at/centos-stream/9-stream/BaseOS/x86_64/os/
enabled=1
gpgcheck=0
EOF

# Clean and verify
dnf repolist
install something
```

4. **Find lines containing ”wild” and save to /root/result**

```
grep wild /usr/share/dict/words > /root/result
```

5. **Find files belonging to user1 and copy to /root/output**

```
mkdir -p /root/output
find / -user user1 -exec cp -r {} /root/output/ \; 2>/dev/null
```

6. ### Compress directories

```
# Compress /usr/tmp to tar.gz
tar -czf /root/compress1.tar.gz /usr/tmp

# Compress /usr/include to tar.bz2
tar -cjf /root/compress2.tar.bz2 /usr/include
```

7. ### Configure cron job running every 2 minutes

```
# Edit crontab
crontab -e

# Add this line:
/2 * * * * logger -p local3.info "wt redhat"
```

8. ### Configure chrony server

```
# Edit chrony configuration
vi /etc/chrony.conf

# Add or replace server line:
pool 154.70.152.179 iburst

# Restart and enable chrony
systemctl restart chronyd
systemctl enable chronyd

# Verify
chronyc sources
```

9. ### Create user anna with autofs

```
# On Node1 (server):
useradd -u 5003 -d /node1/anna anna
mkdir -p /node1/anna
chown anna:anna /node1/anna

# Install and configure NFS
dnf install nfs-utils -y
echo "/node1/anna 172.24.40.0/24(rw,sync,no_root_squash)" >> /etc/exports
systemctl enable --now nfs-server
exportfs -av

# On Node2 (client):
useradd -u 5003 anna
dnf install autofs nfs-utils -y

# Configure autofs
echo "/node2 /etc/auto.node2" >> /etc/auto.master
echo "anna -fstype=nfs,rw 172.24.40.40:/node1/anna" > /etc/auto.node2

systemctl enable --now autofs
```

10. **Host web server on port 82**

```
# Install httpd
dnf install httpd -y

# Create files
echo "Content of file1" > /var/www/html/file1
echo "Content of file2" > /var/www/html/file2
echo "Content of file3" > /var/www/html/file3

# Configure port 82
sed -i 's/Listen 80/Listen 82/' /etc/httpd/conf/httpd.conf


sed -i 's/DirectoryIndex index.html/DirectoryIndex file1 file2 file3/'
/etc/httpd/conf/httpd.conf
# Configure SELinux for port 82
semanage port -a -t http_port_t -p tcp 82

# Configure firewall
firewall-cmd --permanent --add-port=82/tcp
firewall-cmd --reload

# Start and enable httpd
systemctl enable --now httpd
```

11. **Create users: Susan, antoine, julien**

```
# Create group linux
groupadd linux

# Create Susan
useradd -s /bin/sh -u 1500 -G linux Susan

# Create antoine (non-interactive)
useradd -s /sbin/nologin antoine

# Create julien with base directory /dir
mkdir -p /dir
useradd -b /dir julien
```

12. **Set default permissions for julien**

```
# Files:  rw- -r-- --- (640)
# Directories:  rwx r-x --- (750)
# umask = 027

# Add to julien's .bashrc or .bash_profile
echo "umask 027" >> /dir/julien/.bashrc
```

13. **Create collaborative directory /common/share**

```
# Create group
groupadd sysadms

# Create directory
mkdir -p /common/share
chown root:sysadms /common/share
chmod 2770 /common/share

# The 2 sets SGID bit so new files inherit sysadms group
```

14. **Create user dbuser1 and build pdfconvert image**

```
# Create user
useradd dbuser1
passwd dbuser1

# Create host directories
mkdir -p /action/incoming /action/outcoming
chown -R dbuuser1:dbuser1 /action
chmod -R 777 /action

# Install podman/docker
dnf install podman -y

# Switch to dbuser1
ssh dbuser1@localhost

# Create working directory
mkdir pdfconvert && cd pdfconvert

# Download Dockerfile and application
wget https://raw.githubusercontent.com/sachinyadav3496/Text-ToPDF/
master/Dockerfile
wget https://raw.githubusercontent.com/sachinyadav3496/Text-ToPDF/
master/pdf_converter.py

# Build image
podman build -t pdfconvert .
```

15. ## Create container with volumes

```
# Run container as dbuser1
podman run -d --name pdfconvertor \
-v /action/incoming:/data/input:Z \
-v /action/outcoming:/data/output:Z \
pdfconvert
```

16. ## Create systemd service for container

```
mkdir -p ~/.config/systemd/user/
cd ~/.config/systemd/user/
# As dbuser1, generate systemd unit file
podman generate systemd --name pdfconvertor --files --new

vim ~/.config/systemd/user/container-pdfconvertor.service

edit restart=always wantedby= ...  add multi-user.targe

# Enable lingering for dbuser1 (as root)
loginctl enable-linger dbuser1

# As dbuser1, enable and start service
systemctl --user daemon-reload
systemctl --user enable container-pdfconvertor.service
systemctl --user start container-pdfconvertor.service
# Or use:
systemctl --user enable --now container-pdfconvertor.service
#reboot and make sure the container works fine
```

2 Node2 Configuration Solutions

1. **Set root password to ”alberate”**

```
At the GRUB boot menu, use the arrow keys to select the boot entry.
Press e to edit the boot parameters
Locate the line starting with linux , which ends with rgb quiet
Add the following at the end of the line:
# init=/bin/bash
Press Ctrl + X to boot into a minimal shell.

Remount the Filesystem as Read-Write:
# mount-o remount,rw /

Reset the Root Password:
# passwd root

Relabel SELinux:
# touch /.autorelabel

Reboot the System:
# exec /usr/lib/systemd/systemd
```

2. **Configure Host Name, IP Address, Gateway and DNS**

```
# Set hostname node2.tekup.com
hostnamectl set-hostname node2.tekup.com

# Configure network(using nmcli) , this is the hard and not recommended way of
configuring the network
nmcli con mod eth0 ipv4.addresses 172.24.40.41/24
nmcli con mod eth0 ipv4.gateway 172.24.40.1
nmcli con mod eth0 ipv4.dns 172.24.40.1
nmcli con mod eth0 ipv4.method manual
nmcli con up eth0

# Configure network and hostname using ***** nmtui ******
```

3. **Configure repos AppStream and BaseOS**

```
# Same as Node1
cat > /etc/yum.repos.d/appstream.repo << EOF
[appstream]
name=AppStream
baseurl=https://centos.anexia.at/centos-stream/9-stream/AppStream/x86_64/os/
enabled=1
gpgcheck=0
EOF

cat > /etc/yum.repos.d/baseos.repo << EOF
[baseos]
name=BaseOS
baseurl=https://centos.anexia.at/centos-stream/9-stream/BaseOS/x86_64/os/
enabled=1
gpgcheck=0
EOF

dnf repolist
install something
```

4. **Create logical volume ”data” with 81 PEs**

```
# Create physical volume (assuming /dev/sdb)
pvcreate /dev/sdb

# Create volume group with 16MB PE size
vgcreate -s 16M storage /dev/sdb

# Create logical volume with 81 PEs
lvcreate -l 81 -n data storage

# Format with XFS
mkfs.xfs /dev/storage/data

# Create mount point and mount
mkdir -p /data
mount /dev/storage/data /data

# Make permanent
echo "/dev/storage/data /data xfs defaults 0 0" >> /etc/fstab
```

5. **Extend data LV to 1300M-1316M**

```
# Current size:  81 PEs * 16MB = 1296MB
# Need to extend to 1300M-1316M (approximately 1 more PE)

lvextend -r -l +1 /dev/storage/data

# Or use:
lvextend -l +1 /dev/storage/data

xfs_growfs /data # Resize XFS filesystem
```

6. **Add 512MB swap partition permanently**

```
# Create swap logical volume
lvcreate -L 512M -n swap storage

# Format as swap
mkswap /dev/storage/swap

# Enable swap
swapon /dev/storage/swap

# Make permanent
echo "/dev/storage/swap none swap defaults 0 0" >> /etc/fstab

# Verify
lsblk free -mh
```

7. **Set recommended tuned profile**

```
# Install tuned if not present
dnf install tuned -y

# Start and enable tuned
systemctl enable --now tuned

# Get recommended profile
tuned-adm recommend

# Set the recommended profile (usually "virtual-guest" or "throughput-performance")
tuned-adm profile $(tuned-adm recommend)

# Verify
tuned-adm active
```

Important Verification Steps

- Test network connectivity: `ping 172.24.40.1`
- Verify services are running: `systemctl status <service>`
- Check firewall rules: `firewall-cmd --list-all`
- Verify mounts persist after reboot
- Test user permissions and autofs functionality